



infinITY Online Camp JEE (Adv) 2024

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SUBJECT: PHYSICS

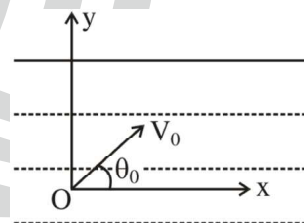
DATE: 15-03-2024

SINGLE CORRECT TYPE (+3, -1)

1. On a road going out of a city, lights of the last traffic signal glow green for 1.0 min and red for 2.0 min. After the traffic signal, the road is straight and vehicles run at their constant speeds that lie in the range 60 km/h to 80 km/h . This mode of operation of traffic lights causes the vehicles to come out of the city in groups. Some distance away from the traffic signal, these groups merge and become indistinguishable. Assuming that the vehicles acquire their constant speeds after the traffic signal in negligible time, which of the following conclusions can you make?

- (A) The groups become indistinguishable 4.0 km after the traffic signal.
- (B) The groups become indistinguishable 8.0 km after the traffic signal.
- (C) Wider is the range of speed of the vehicles; greater is the distance where the groups become indistinguishable.
- (D) Larger is the number of fast moving vehicles, shorter is the distance where the groups become indistinguishable.

2. A projectile of mass m is fired at an angle θ_0 , with horizontal and initial speed V_0 , as shown. If the air develops a drag force (resistance force) on the projectile which is proportional to its instantaneous velocity i.e. $F = -KV$ where K is a positive constant of appropriate dimensions. (Assume that gravitational acceleration is constant throughout the motion i.e. gm/s^2)



- (A) The time of ascent is greater than the time of descent.
- (B) Vertical (Y-component) component of velocity of projectile as a function of time is

$$\frac{m}{K} \left[\left(\frac{K}{m} V_0 \sin \theta_0 - g \right) e^{\frac{-K}{m}t} + g \right]$$

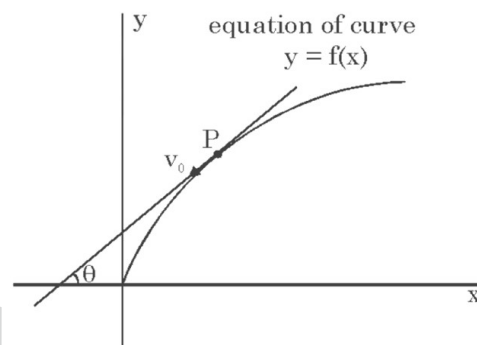
- (C) Horizontal (X-component) component of velocity of projectile as a function of time is $V_0 \cos \theta_0 e^{\frac{K}{m}t}$
- (D) Vertical (Y-component) component of velocity of projectile as a function of time is

$$\frac{m}{K} \left[\left(\frac{K}{m} V_0 \sin \theta_0 + g \right) e^{\frac{-2K}{m}t} + g \right]$$

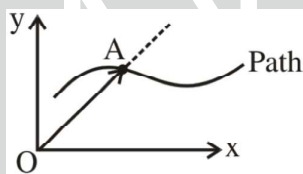
ONE OR MORE THAN ONE OPTION(S) CORRECT: (+4, -1)

3. A small particle is moving in $x-y$ plane on a smooth wire bent in curve shape as shown in figure. At a certain instant it is at P and moving with speed v_0 . Here gravity is in negative y -direction and mass of particle is m . At this instant y -component of acceleration is $-6\hat{j} m/s^2$.

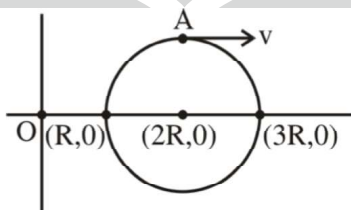
Given $v_0 = 2 m/s, m = 2 kg, \theta = 45^\circ$ then :-



- (A) x -component of acceleration is $-6\hat{i} m/s^2$.
- (B) Radius of curvature at P is $2\sqrt{2} m$.
- (C) At P , x -component of acceleration is $-4\hat{i} m/s^2$.
- (D) At P , normal reaction between particle and wire is $8\sqrt{2} N$.
4. When a particle is in motion along a curved path, its velocity at a point A is along the tangent to the path at point A as shown in the first figure. We may split this velocity into two components; one along position vector OA and call it radial velocity and another along the perpendicular to OA and call it transverse velocity. Same can be done to acceleration at A and call the components radial acceleration and transverse acceleration.



A particle is executing uniform circular motion of angular velocity ω in $x-y$ plane with radius R center $(2R, 0)$. When it is at point A , answer the correct option(s).



- (A) Ratio of angular velocity to the radial velocity of the particle about origin is $1/R$.
- (B) Ratio of angular velocity to the radial velocity of the particle about origin is $\frac{1}{2\sqrt{5}R}$.
- (C) Net acceleration of particle about origin is $R\omega^2$.
- (D) To find angular acceleration about the origin in the given question we can apply $a_t = r\alpha$, where a_t is transverse acceleration about point O and r is distance between point O and A .

5. A river of width ω is flowing such that the stream velocity varies with y as $v_R = v_0 \left[1 + \frac{\sqrt{3}-1}{\omega} y \right]$ where y is the perpendicular distance from one bank. A boat starts rowing from the bank with constant velocity $v = 2v_0$ in such a way that it always moves along a straight line perpendicular to the banks with respect to ground.

(A) The time he will reach the other bank is $\frac{\omega\pi}{3(\sqrt{3}+1)v_0}$

(B) The time he will reach the other bank is $\frac{\omega\pi}{6(\sqrt{3}-1)v_0}$

(C) The velocity of the boat along the straight line when he reaches the other bank is v_0

(D) The velocity of the boat along the straight line when he reaches the other bank is $3v_0$

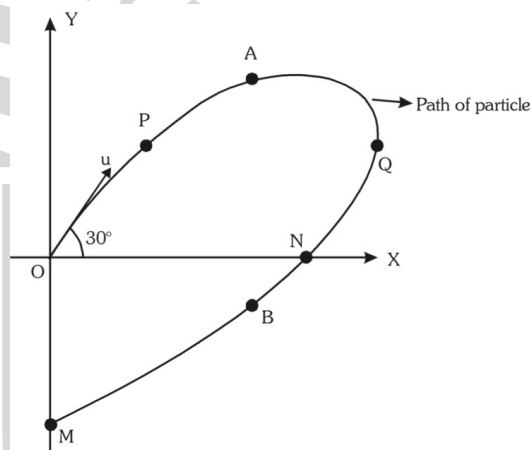
6. A particle is projected from origin with speed $u = 50 \text{ m/s}$ at an angle 30° with X -axis. Due to wind particle acquire an extra acceleration $a = 6 \text{ m/s}^2$ in $-ve$ x -direction. Path of particle is shown in figure.

x - Coordinates of point A and B are same

y - Co-ordinates of P and Q are same.

Point N on the X -axis, point M on the Y -axis.

Time taken by particle from O to P , O to Q , O to A and O to B are t_1, t_2, t_3 and t_4 respectively. Then match the column. (Assuming Y along vertical and there is no surface in the space) ($g = 10 \text{ m/s}^2$)



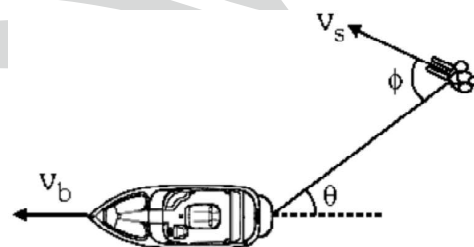
(A) $t_1 + t_2$ is 5 sec

(B) $t_3 + t_4$ is $\frac{25}{\sqrt{3}}$ sec

(C) length ON is $25\sqrt{3}(5 - \sqrt{3}) \text{ m}$

(D) length ON is 5 m

7. A boat is moving with uniform velocity $v_b = 20 \text{ m/s}$ pulling a water skier with the help of a tug-rope of length $l = 10 \text{ m}$. To increase his speed the water skier tilts the skis slightly away from the direction of motion of the boat. As he does so, the tug rope rotates.



(A) The speed v_s of the skier with respect to the ground is $20\sqrt{3} \text{ m/s}$. (when $\theta = 30^\circ$ and $\phi = 60^\circ$)

(B) The speed v_s of the skier with respect to the ground is $30\sqrt{7} \text{ m/s}$. (when $\theta = 30^\circ$ and $\phi = 60^\circ$)

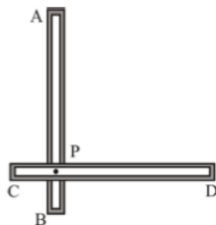
(C) Angular velocity ω of the rope is 1 rad./s (when $\theta = 30^\circ$ and $\phi = 60^\circ$)

(D) Angular velocity ω of the rope is 2 rad./s (when $\theta = 30^\circ$ and $\phi = 60^\circ$)

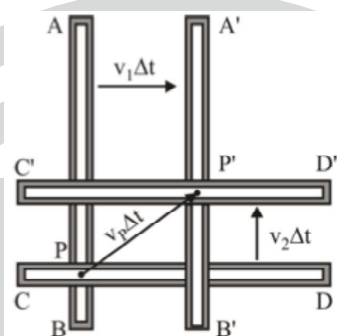
PARAGRAPH BASED QUESTION(S) EACH CORRECT: (+3, -1)

Paragraph - 1 for Question Nos. 8 to 9

When two rigid bodies move over each other, their intersection point might not be moving with the velocity of either of them. For example, consider two rods AB and CD , connected by pin P that moves in slots made in the rods.

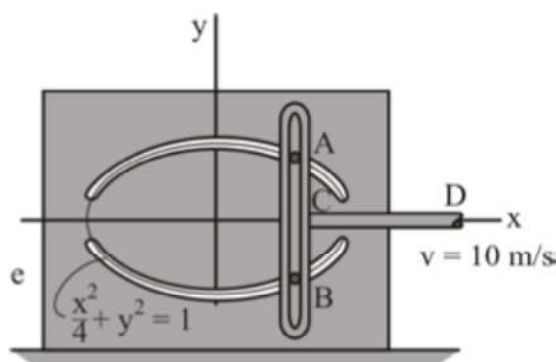


If the rod AB is moving with velocity v_1 and rod CD is moving with velocity v_2 , their intersection point (pin P) moves as shown:



Where $A'B'$ and $C'D'$ are the new positions of the rods, while P' is the new position of the pin P . Here the pin P is having a velocity that is the sum of the velocities of the two rods. But in more complicated shapes of the sliders, this direct approach might not work, and calculus is required to calculate the velocity or acceleration of the intersection point.

In the given diagram, there is an elliptical slot as shown. The equation of the ellipse is $\frac{x^2}{4} + y^2 = 1$, where x and y are the coordinates in same unit. There is a slider with two pins A and B , which are free to move along the slider so that they always remain on the elliptical slot. The slider is moved horizontally with constant speed 10 m/s . Answer the following questions.



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8. When x -coordinate of the slider is 1 m , the speed of Pin-B is?
(A) Zero (B) 10 m/s (C) 2.9 m/s (D) 10.4 m/s
9. When x -coordinate of the slider is 1 m , the magnitude of acceleration of Pin-B is?
(A) Zero (B) 38.4 cm/s^2 (C) 38.5 m/s^2 (D) 42 cm/s^2

NUMERICALS TYPE: (+4, -1)

10. A particle projected from the ground passes two points, which are at heights $h_1 = 12\text{ m}$ and $h_2 = 18\text{ m}$ above the ground and a distance $d = 10\text{ m}$ apart. What could be the minimum speed of projection? Acceleration due to gravity is $g = 10\text{ m/s}^2$.

